Bounded Agency: Ontology-Driven Authority for Delegated Intelligence

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Abstract

An autonomous system's technical permissions may let it perform actions that settle decisions beyond the authority it was given. Permission to update an incident record, for example, need not include authority to declare a potentially hazardous situation resolved. Ontology-grounded policies can express these decision boundaries, but revisions to classifications, relationships, or inference rules can change what an unchanged policy permits. This paper examines bounded agency as delegated independent judgment limited by the situations an agent may resolve, the outcomes it may choose, and the consequences it may initiate. Its central mechanism is authority-relative semantic-change admission: assessing how a proposed change in meaning affects a specified agent's decision scope before approving that change for operational use. A lightweight model separates the decisions a candidate semantic model would permit in a hypothetical evaluation from the decisions the agent is currently authorized to carry out. A positive-taxonomy counterexample shows how a subclass axiom enlarges the former set without editing policy text. A reference architecture connects that comparison to explicit approval, version-bound operational authorization, and controlled execution. Building-safety and customer-support thought experiments distinguish changes in meaning from changes in accepted facts. The contribution is a conceptual design synthesis building on semantic-policy and ontology-evolution research. The cases and finite taxonomy illustration explain the mechanism; operational effectiveness remains to be evaluated.

Keywords: bounded agency, delegated intelligence, decision authority, ontology evolution, semantic policy, agent governance, authority-sensitive change

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1. Introduction

Consider a facilities agent that monitors building incidents. It may record observations, notify a designated responder, and close independently verified routine maintenance notices. Its technical permissions allow it to update incident status. The same interface can mark an unexplained sensor alert as `resolved`; in this hypothetical application, that update removes the alert from the unresolved-incident display and may send an all-clear message to occupants. The agent can invoke the operation, but it has not thereby been given authority to decide that a potentially hazardous situation is safe.

The permissions granted by the software can therefore exceed the boundary of the decisions entrusted to the agent. That boundary concerns which situations it may independently resolve, which outcomes it may choose, and which consequences it may initiate. It must still be able to notice unfamiliar or potentially dangerous situations, preserve observations, and escalate under its reporting authority. A limit on autonomous resolution is not a reason to ignore a hazard.

This is a configuration and modeling problem rather than evidence that authorization systems are intrinsically limited to identity and resource access. Attribute-based, relationship-based, and usage-control systems can express rich business conditions [4-6]. A well-designed authorization policy may already enforce the decision boundary. The issue is whether the deployment represents and enforces that boundary, including evidence requirements and downstream effects. Broad access to an update operation does not supply those distinctions automatically.

A shared ontology can support them. Domain concepts such as incident, observation, routine notice, unresolved alert, responder, and resolution provide a common language for relating a situation to an authorized decision. Yet that language itself can change. Suppose a policy permits an agent to close routine notices. A revised ontology places a kind of sensor anomaly not previously classified as routine within that category. Where routine membership defines the settlement scope, the same policy can now permit additional closures. If a deployment adopts that revision without grant review, the technical permissions and policy text remain unchanged while the situations the agent can resolve expand. Section 7 distinguishes this case from one in which an independent safety-clearance requirement still prevents closure.

The question examined here is: **How can an autonomous intelligence be given a bounded scope of independent judgment, grounded in a shared model of its operating world, without allowing learning or semantic evolution to silently expand its authority?** The word *silently* matters. An authorized person or organization may intentionally

broaden the agent's delegated decision scope. The design objective is to make a relevant expansion detectable, reviewable, and explicitly admitted, rather than to freeze knowledge or prohibit institutional change.

We call this constrained scope of independent judgment *bounded agency*. A *mandate* means the assignment that specifies which situations, decisions, and consequences have been entrusted to the agent. The assignment may come from a person or an organization. It is distinct from the technical permissions through which the agent can affect a system.

The paper's central mechanism is **authority-relative semantic-change admission**. In plain terms, it asks: does a proposed change in meaning expand what this agent would be allowed to decide, and who may approve using that change? *Authority-relative* identifies the particular agent, mandate, and policy being examined. *Semantic change* includes revisions to classifications, relationships, mappings, or inference rules. *Admission* is the decision to accept the revised model as a basis for operational decisions. This change-approval process is distinct from authorizing an individual action under the currently approved model.

The paper makes three contributions. First, it gives an explicit comparison target for changes to the effective permissible-decision set under a fixed policy and controlled context. Second, it connects that comparison to a proposed semantic-change admission procedure and a version-bound authorization for each proposed decision. Third, it develops two operational thought experiments that distinguish ontology revisions from ordinary facts and classification assertions. These contributions are architectural proposals and explanatory formalization. They build on established authorization and semantic-change mechanisms; their operational benefits require evaluation.

2. Related work

2.1. Approach and scope of comparison

This conceptual study formulates a change-control problem, defines a comparison target, establishes a restricted counterexample, and traces the proposed controls through two synthetic cases. The analytical criteria are whether the comparison isolates semantic change, whether candidate eligibility remains separate from operational authorization, and whether the proposed approval roles can prevent a candidate from authorizing its own adoption. These checks assess coherence and explanatory value; they do not measure effectiveness.

The literature comparison draws on targeted searches of primary papers, official specifications and documentation, and references from closely related work, with focused rechecking on 15 October 2026. Coverage is selective

rather than systematic. Recent preprints and live specifications are discussed as proposals, not independently validated systems. The bibliography identifies the versions examined. For the enterprise framework [14], the institutional abstract and publisher-indexed passages were accessible, but complete direct retrieval was limited.

2.2. Semantic policy and institutional meaning

KAoS provides ontology-based policy specification, analysis, management, and enforcement for agents and distributed services [1, 23]. Rei connects policy rules to domain ontologies and supports permissions, obligations, delegation, and revocation [2, 28]. These frameworks establish ontology-grounded agent-policy specification, management, and enforcement.

Normative multiagent-system research supplies a closely related conceptual foundation. Boella and van der Torre distinguish regulative norms from constitutive rules that determine what counts as an institutional fact [3]. Classification can therefore carry normative consequences, and authority over institutional rules is not reducible to an agent's ability to act. Jones and Sergot also distinguish institutional power from permission and practical possibility [25]. The category-expansion example applies this established relationship to semantic-change admission.

ODRL specifies permissions, prohibitions, duties, constraints, and parties [9]. OWL supplies formal ontology constructs and entailment semantics [10]; SHACL supplies graph validation against shapes [11]. None of those specifications alone establishes the legitimacy of a business decision. They are usable components whose normative significance depends on the institution's policy and implementation.

2.3. Authorization, delegation, and enforceable behavior

ABAC evaluates attributes of subjects, objects, operations, and environment [4]; Zanzibar demonstrates relationship-based authorization at scale [5]; UCON integrates authorizations, obligations, conditions, and mutable attributes with ongoing usage control [6]. Cedar provides a policy language, evaluator, and formal analysis facilities [7]. OPA separates policy decisions from application code [8]. Thus independent decision evaluation and rich contextual authority conditions are established capabilities.

The decision-rights literature separates management of a decision from its ratification and monitoring [21]. This supports treating evidence gathering, proposing an outcome, approving it, and executing it as different assignments. It does not imply that an agent becomes the legal bearer of accountability when it receives a software role.

Runtime enforcement is also established. Schneider's analysis concerns the security properties enforceable by execution monitors [20]. Our enforcement claims are correspondingly limited to represented, mediated operational behavior. An evaluator cannot inspect or prohibit all private inference, nor can a gate prevent effects that bypass it.

2.4. Semantic evolution and recent governed agents

Ontology-evolution research already investigates preservation and change impact. Conservative extensions and inseparability provide formal ways to ask whether relevant consequences or queries change [12]. Security coevolution work uses ontology change patterns to detect security impact and propose adaptations [13]. The effective-authority comparison proposed here is therefore an application-specific target for established change-analysis techniques, rather than a new general theory of semantic difference. Lopes and colleagues already propagate permission annotations across RDF inference premises [24], illustrating why classification and its evidential basis belong in the comparison.

Recent work addresses related governance boundaries. Jørgensen and Ma combine enterprise semantics, contextual knowledge, process orchestration, decision authority, execution control, and provenance in a bounded-autonomy framework [14]. AgentBound separates behavioral and semantic constraints and binds action governance to versioned artifacts [15]. The Authority Resolution Framework models contextual authority and provenance through an ontology [16]. Agent libOS separates evolving action surfaces from admission authority [17], and AgentFlow addresses delegated task scope, runtime monitoring, and bounded policy verification [18]. Salas specifies commit-time authority checks over changing normative and model dependencies [19]. These works overlap with important parts of the proposed architecture, although they differ in emphasis and evidence.

An undated live artifact-evolution specification also separates proposals, qualified approval, immutable versions, realization, and declared semantic comparison coverage [22]. Its public text is relevant overlap, but the review does not establish its historical priority.

The focus here is an explicit permissible-decision-set comparison and its connection to mandate ratification, including the distinction between ontology revisions and fact admission. The paper develops a synthesis and specialization of the mechanisms above; it makes no claim of priority for ontology-based agent governance or for the combination as a whole.

2.5. The specific design contribution

For a policy expressed through ontology queries, the expansion comparison is a policy-selected query-answer difference, closely related to the notions surveyed in [12]. The additional design question is who may adopt a changed answer set as the basis of a particular delegation. Semantic difference computation supplies evidence for that decision; it does not supply the right to approve it.

The broader architecture also has substantial antecedents. KAoS combines ontology-based policy reasoning and enforcement [23]; Jørgensen and Ma connect enterprise semantics with delegated authority and controlled execution [14]; AgentBound binds action judgments to delegation, policy, and semantic artifacts [15]; and Salas checks current authority and model dependencies at commitment [19]. Those capabilities are foundations for this proposal. Here the organizing object is a proposed semantic release: compare its candidate decision set before adoption, distinguish that comparison from current authorization, and route the result to the role entitled to revise the grant. Existing systems could implement this procedure. Its value therefore depends on whether making that obligation explicit improves change review, a question the present illustration cannot answer.

3. Delegation and the dimensions of bounded agency

3.1. Four distinctions

Delegated intelligence emphasizes the relationship between a system and the human or institution entrusting it with work. It complements rather than replaces a technical description of artificial intelligence. A delegation can assign a problem, access, interpretation, choice among outcomes, execution, investigation, or adaptation. Those assignments need not travel together. Authenticated delegation and intelligent-delegation research already addresses task scope and the transfer of authority and responsibility [26, 27].

We distinguish four operational concepts:

1. **Capability:** the computation or operation the system can technically perform. Capability does not establish reliable competence.
2. **Execution permission:** authorization to invoke an operation on a resource. A sophisticated implementation may already condition this permission on business meaning.
3. **Decision authority:** the delegated right to resolve specified kinds of situations, select or approve outcomes, and initiate their consequences under stated conditions.

4. **Semantic authority:** a right to promote a representation or inference regime into the authoritative basis of operational decisions.

Semantic authority is preferable here to *epistemic authority*. The latter can refer to credibility, expertise, or entitlement to believe. Our concern here is whether an ontology axiom, mapping, classifier output, or evidence-admission rule is approved for operational use. This is distinct from assessing which claims the resulting evidence policy warrants accepting. Authority to adopt that policy does not establish that a source is truthful or that its reports are independent. Approval does not make it true. Similarly, a passing policy evaluation is an implementation's representation of a mandate, not proof of ethical or legal legitimacy.

Responsibility and accountability remain distinct from these permissions. An agent may be assigned to produce a recommendation; a designated institutional role may remain responsible for ratifying it and answering for its consequences. The architecture must identify that role rather than rely on a statement that the agent is accountable.

3.2. Scope, constraints, and admitted semantic basis

The first dimension is **decision scope**: the situations, subjects, and decision classes entrusted to the agent. Recording a sensor observation, closing a routine maintenance notice, and declaring an unresolved situation safe are different decisions even when they concern the same location.

The second is **decision constraints**: permitted outcomes, conditions, and consequences. An agent may recommend an assessment while lacking authority to issue an all-clear. Location, recipient, evidence freshness, required review, and cumulative effects may constrain an otherwise permitted decision class.

The third is **the admitted semantic basis**: which formal representations and admission rules may support those operational judgments. An agent can recognize a possible new category, preserve evidence, and propose a model change without being allowed to treat the proposal as an approved ground for action.

These dimensions interact. A taxonomy change affects scope; a changed relationship may affect an outcome condition; a new assertion can affect both without revising the ontology. They are analytical distinctions, not independent mathematical axes.

A frontier model's latent knowledge is broader than a domain ontology. We do not claim to remove that knowledge from its weights or prevent internal speculation. The enforceable boundary concerns represented proposals,

admissible evidence and classifications, adopted model changes, and operational effects. Out-of-scope analysis can itself have consequences when it discloses data or produces an influential institutional record; those outputs must be represented as effects if they are to be governed.

4. A lightweight model of effective authority

4.1. Separate objects

Let A identify an agent acting through an authenticated role and delegation chain. Let O_v be a versioned domain ontology, K a situation knowledge base, and E evidence supporting assertions in K. Observations enter through an admission process; they are not automatically authoritative facts.

Let P be the substantive decision-authority policy, L the substantive delegation mandate, and C the operational context, including relevant time, jurisdiction, workflow state, revocation, and previously committed consequences. Let a **semantic bundle** be

$$B = (O_v, I, M, R, T),$$

where I identifies imports, M mappings and entity-resolution rules, R the inference regime, and T assertion-admission rules, including validators, shape versions, and classifier artifacts. A manifest identifies their immutable versions or digests. The bundle is broader than the ontology because policy interpretation can change through any of these inputs. The deployment manifest additionally identifies evaluator semantics and implementation, effect-contract versions, and trusted adapter versions; changes to these artifacts enter the same admission and freshness boundary.

A proposal is

$$q = (d, x, y, h),$$

where d is the decision class, x the governed subject or subjects, y the proposed outcome, and h a represented effect plan. The effect plan identifies intended operations and declared consequence dependencies. It is not an oracle for every real-world consequence.

Define a **counterfactual eligibility evaluator**

$$U(A, q; P, L, B, K, E, C) \in \{\mathsf{permit}, \mathsf{deny}, \mathsf{indeterminate}\}.$$

It checks substantive policy and mandate conditions, supported situation assertions, and represented consequences under the supplied bundle. For comparison, it intentionally excludes the requirement that the candidate bundle

already be admitted by an active production grant. It runs in a hypothetical, or shadow, evaluation with no authority to cause effects. Indeterminate covers required conditions that remain unresolved under the declared evidence and conflict policy, including missing evidence, unsupported inference, or unknown classification. A conflict does not automatically make every decision indeterminate: unaffected facts can still establish a refusal or permission. If the policy requires warrant across all admissible consistent interpretations, permission requires the release conditions to hold in every one; an empty set of interpretations must never authorize by vacuous universal quantification. Deployment rules must specify these choices; unrestricted classical entailment from an inconsistent theory is unsuitable for a permit test.

For a fixed proposal universe Q, define the candidate eligibility envelope

$$\widehat{\mathcal{H}}_Q(A; P, L, B, K, E, C) = \{q \in Q \mid U(A, q; P, L, B, K, E, C) = \mathsf{permit}\}.$$

Let G identify the active version-bound grant. **Operational authorization** V determines whether the agent may actually carry out this particular proposal now. It additionally requires that G admit this exact policy, mandate, semantic and deployment basis; that the grant and evidence remain current; and that the effect can be faithfully mediated. Thus $U = \mathsf{permit}$ is necessary but not sufficient for $V = \mathsf{permit}$. A candidate B_1 can enlarge $\widehat{\mathcal{H}}_Q$ while remaining inadmissible for production under G_0.

The operational authority envelope contains only proposals permitted by V under an active grant. Separating it from the candidate envelope avoids a circular change test: if the comparison already demanded ratification of the candidate, it would conceal the very prospective expansion being reviewed. Neither envelope establishes substantive correctness or institutional legitimacy.

4.2. Authority-relative change

Hold A, P, L, K, E, C, Q fixed and compare bundles B_0 and B_1. This comparison also holds evaluator and effect-contract semantics fixed. Changes to evaluator, adapter, or effect-contract versions require a separately declared comparison or revalidation of those deployment artifacts. Define the expansion witnesses

$$\Delta^+_Q(B_0, B_1) = \widehat{\mathcal{H}}_Q(A; P, L, B_1, K, E, C) \setminus \widehat{\mathcal{H}}_Q(A; P, L, B_0, K, E, C).$$

A nonempty set exhibits expansion under the controlled comparison. An indeterminate proposal under the old basis becoming permitted under the

new one counts as expansion. A removal set can likewise be defined by reversing the difference; lost permissions can require review even without expansion.

This definition separates semantic effects from ordinary changes in the situation. In production, K, E, and C also vary. Controlled replay can compare the same snapshot to isolate the bundle's impact, while operational authorization evaluates the current admitted context. Holding already-admitted K fixed tests downstream interpretation only. To compare changes to T or classifiers, replay the same original observations and evidence through both admission pipelines and compare their resulting situation models; this is a distinct controlled comparison. A newly eligible customer under an unchanged mandate is not necessarily a new delegation; it may be an intended application of the existing rule. A falsely admitted classification can nevertheless create an unauthorized effect. The controls on T, K, and E therefore matter independently of ontology release controls.

A stronger **non-expansion** obligation would quantify over an explicitly chosen comparison family $\mathcal{W}$ of situation/evidence/context triples chosen independently of either candidate's acceptance rules:

$$\forall (K, E, C) \in \mathcal{W}, \quad \widehat{\mathcal{H}}_Q(A; P, L, B_1, K, E, C) \subseteq \widehat{\mathcal{H}}_Q(A; P, L, B_0, K, E, C).$$

This inclusion forbids new permits but allows existing permits to disappear. **Permit-set equivalence** instead requires equality over the same comparison family, or both inclusions. Neither property establishes preservation of obligations, denial reasons, or other behavior omitted from the permit set. In particular, a deny-to-indeterminate transition changes the evaluator's result without changing eligibility. A review that depends on those distinctions must retain the full outcome and any associated obligations. A revision can also add some permits and remove others; both findings must be reported.

Both Q and $\mathcal{W}$ are fixed before comparison, over declared relevant input and decision signatures, including any proposed new vocabulary. That statement names the scope of the obligation; this paper supplies no general algorithm proving it. A finite replay corpus establishes non-expansion only over that corpus. Conservative extension over an old ontology signature is not automatically preservation over new decision vocabulary, future instances, mappings, and policy queries. The relevant signatures, data domain, and query language must be specified.

4.3. A restricted counterexample

Consider finite class names and individuals, positive subclass axioms, positive type assertions, and transitive classification. Fix one policy permitting

closure of a notice x exactly when $K \cup O$ entails `RoutineNotice(x)` and the unchanged evidence/context predicates pass. There are no overriding denials, negation, inconsistent assertions, or other inference rules.

Proposition. For counterfactual eligibility in this fragment, adding subclass axioms cannot remove a previously permitted closure. It can add a closure without changing the policy.

Proof. Every existing derivation of `RoutineNotice(x)` remains available after axioms are added, and the other predicates are fixed. Therefore the permitted set cannot shrink. For strict growth, choose K and O_0 such that K contains `SensorAnomaly(c)`, the unchanged evidence/context predicates pass, and $K \cup O_0$ does not entail `RoutineNotice(c)`. Under the evaluator's stipulated positive-membership permit rule, closure of c is not permitted. Add `SensorAnomaly` subclass of `RoutineNotice` to obtain O_1. The new axiom derives `RoutineNotice(c)`, so closure is permitted. The policy is identical. □

This is a direct consequence of monotone positive inference, not a new mathematical theorem. Non-entailment is not an OWL assertion of nonmembership; the gate simply refuses to permit without the required entailment. Deny precedence, negative tests, changed evidence, inconsistent ontologies, or nonmonotonic rules invalidate the general monotonicity claim. The companion executable example demonstrates only this finite fragment.

4.4. Executable illustration

The supplied program enumerates four fixed incident-closure proposals. Under B_0, service notices are routine; B_1 adds the sensor-anomaly subclass axiom. Unless noted, proposals concern the assigned site and have attested observations. The table reports Boolean eligibility, where false means nonpermit and does not distinguish denial from indeterminacy.

Fixed proposal	Eligible under B_0	Eligible under B_1
Routine service notice	True	True
Sensor anomaly	False	True
Sensor anomaly, other site	False	False
Sensor anomaly, unattested observations	False	False

The program finds one expansion witness and no removal. A grant still bound to B_0 refuses the candidate basis; a grant bound to B_1 permits the eligible anomaly in the toy model, and revocation blocks it. An unrelated subclass addition leaves eligibility unchanged. Removing the service-to-routine axiom alone removes the original permit without adding one, illustrating nonexpansion without equivalence. Replacing that axiom with the anomaly-to-routine axiom both removes the service permit and adds the anomaly permit.

These controls are recorded in the example output. A further composition control splits the anomaly-to-routine path into two edits: `SensorAnomaly` becomes a subclass of `ReviewCategory`, and `ReviewCategory` becomes a subclass of `RoutineNotice`. Alone, the first edit supplies only the intermediate category, while the second has no intermediate category to use; neither changes eligibility on the four proposals. Together they add the anomaly closure. Thus separate non-expansion checks do not establish non-expansion of their combination. They establish the stated results for the enumerated proposals, not general comparison coverage or safety of incident closure. The program omits full three-valued evaluation, obligations, evidence authentication, and production execution.

5. Proposed reference architecture

5.1. Components and trust boundaries

The architecture has six components, with traditional IAM retaining identity, authentication, resource access, and authenticated delegation-chain responsibilities.

Component	Responsibility	Output used by the next boundary
Semantic service	Serve admitted ontology, imports, mappings, inference regime, and assertion rules	Immutable bundle manifest and relevant semantic slice
Situation/evidence service	Admit observations, maintain facts, preserve provenance and disputed assertions	Versioned decision snapshot with evidence references
Authority registry	Store mandates, decision policies, approval roles, revocation and consequence rules	Versioned authority context
Decision evaluator	Independently check typed proposals against current authority and supported facts	Permit, deny, or indeterminate with reasons and dependencies
Enforcement adapter	Mediate governed operations and bind actual parameters to admitted effects	Execution or refusal plus effect receipt
Governance process	Review semantic and authority changes, approve releases and rollout	Separate semantic acceptance and authority-ratification records

The semantic service need not be a single physical server or require a universal ontology. A governed domain model can be shared through a knowledge graph or registry. Agents can retrieve small slices, provided the evaluator resolves the authoritative closure and manifest rather than trusting an agent-supplied fragment to be complete. Multiagent retrieval does not imply equal grants: common meaning and delegated authority are separate.

The evaluator should be independently controlled relative to the operating agent. Policies may use decision classes, ontology predicates, state transitions, outcome constraints, evidence conditions, and delegation context. Whether those checks are implemented in Cedar, OPA, a rule engine, or a custom reasoner is a deployment choice. Using a general language model as the final evaluator requires additional evidence of reliability; calling it an independent reviewer is insufficient.

5.2. Decision envelopes and commit

A submitted envelope should name the authenticated role and delegation chain; mandate and policy versions; semantic and deployment manifests, including evaluator, effect-contract, and adapter versions; situation/evidence snapshot; decision class, subject, outcome, and effect plan; evidence dependencies; expiry; and an effect identity for replay and idempotency. These are proposed fields, not a new credential standard.

Before commitment, the enforcement path must confirm that the envelope is still admitted under the current grant and revocation rules. It must bind the permitted outcome to actual adapter parameters. A permission to recommend cannot authorize an adapter call that commits the recommendation as a final decision. Side effects of routing, communications, and disclosure must be included where they are institutionally governed.

A version check alone is not a transaction. A policy, evidence snapshot, or grant may change between evaluation and effect. An implementation needs a transaction, reservation, lease/fence protocol, or other explicitly analyzed mechanism that gives admission and execution defined freshness semantics. Salas already addresses closely related commit-time dependencies [19]; we adopt that engineering requirement without claiming a new concurrency protocol.

Version pinning also needs revocation. An agent cannot indefinitely retain a permissive historical bundle after the institution withdraws its grant. A migration may deliberately keep an old contract active for a defined scope, but that continued use must itself be admitted. Ordinary receipt replay explains what happened; it does not authorize a new action today.

5.3. Proposed invariants and their limits

The design seeks four invariants:

- Every governed effect passes a controlled enforcement path and matches an admitted proposal.
- Every permit identifies its authority and semantic basis and sufficient current evidence.
- A semantic release cannot substitute a new operational basis into a grant without the required authority review.
- An authority-sensitive change produces an explicit decision to reject, retain a defined old scope, or admit a revised grant.

These are design requirements, not established properties of an implemented system. They depend on complete mediation, credential isolation, authenticated versions, policy correctness, adequate dependency discovery, and faithful adapters. A tamper-resistant receipt can support audit while still faithfully recording a mistaken decision. Availability failures must have defined refusal behavior; safety by refusal can impose substantial operational costs.

6. Admission of semantic and knowledge changes

6.1. Different changes require different questions

Learning new facts about familiar concepts is not the same as changing the conceptual scheme. The distinction helps assign review, but it is not a reliable security boundary by itself.

Change	Example	Appropriate control question
Observation or ordinary fact	Authenticated observation received for a known incident	Is the assertion supported, current, and admitted under the existing evidence rule?
New instance	A new maintenance notice of an existing kind	Does normal enrollment and the existing mandate cover this instance?
Classification assertion	An existing alert is now classified as routine	Who can attest this type, and what evidence permits operational reliance?
Relationship or mapping	External issue code mapped to courtesy issue	Does the mapping change policy-relevant entailments or governed subjects?

Change	Example	Appropriate control question
New class or changed definition	Sensor anomaly becomes a routine-notice subclass	Does this alter the decisions or consequences permitted under current mandates?
Imports or inference settings	A reasoner begins using an additional relation rule	Does the admitted closure or evaluation behavior change?
Authority policy or mandate	Agent receives authority to issue an all-clear	Is the delegating role entitled to grant it, and has the change been explicitly ratified?

A factual update can legitimately alter current eligibility while leaving the delegated rule unchanged. It can also improperly launder a disputed classification into a permit. Assertions that supply policy-relevant types or relationships should therefore pass an approved admission rule. They need not receive the same governance ceremony as every ontology revision.

Likewise, a new concept unrelated to any policy dependency may be harmless for a particular agent's authority. The appropriate target is authority impact relative to declared decisions and context, rather than indiscriminate approval of all learning. Impact analysis may be incomplete, so an unrelated-change verdict needs an explicit coverage claim.

6.2. Proposed change-review procedure

The reference procedure is:

1. **Register a proposal.** Preserve observations, evidence, proposed edits, proposer identity, and the immutable baseline. Keep proposed semantics separate from the admitted operational bundle.
2. **Resolve dependencies.** Identify relevant policies, mandates, decision classes, mappings, imports, assertion rules, and effect contracts. Use independently derived dependencies where possible; agent-declared dependencies are hints requiring validation.
3. **Compare under a stated regime.** Check logical consistency and graph conformance, then compare counterfactual eligibility under U using controlled snapshots. State the proposal language, data domain, reasoner, and coverage. Seek expansion witnesses and removed permissions for the complete proposed change. For a sequential rollout, also check each admitted prefix against the relevant state; individually checked edits do not establish preservation of the combined release.

4. **Return a qualified result.** Report expansion and removal witnesses separately, since both can occur. State coverage independently: proof under declared assumptions, exhaustive evaluation of a declared finite domain, incomplete tests, or indeterminate comparison. Distinguish non-expansion from permit-set equivalence. Retain changed non-permit outcomes and obligations when relevant to the review. Do not relabel a successful regression suite as a universal proof.
5. **Obtain the required decisions.** Reviewer eligibility must be resolved under the pre-change admitted authority context; a candidate cannot create its own approver through a new class or relationship. A domain steward may accept a better ontology without possessing authority to broaden an agent's mandate. Where the policy requires it, an independently authorized role must ratify the affected grant. One person may hold both roles, but the two decisions remain explicit.
6. **Release and migrate.** Publish the admitted bundle, policy/grant versions, scope, effective time, expiry, and revocation behavior. Validate or invalidate pending proposals; retain a defined old scope only under an active grant.
7. **Monitor and retain evidence.** Preserve comparisons, approval reasons, admitted envelopes, and effect receipts. Observe unexpected classifications and effects, and route them to investigation.

The policy for approving a semantic release is itself a governed policy. Shadow comparison is read-only with respect to governed effects and cannot issue an execution grant. An operating agent must not be able to alter that rule or attest its own eligibility merely by proposing a new ontology relation. Escalation should not fall back to a permissive mode when the reviewer is unavailable.

6.3. Knowledge growth without automatic authority growth

The statement "an agent may expand its knowledge without automatically expanding its authority" is a design principle consistent with established separation of duties and admission controls. It does not imply that all facts leave the effective permit set unchanged. Facts are supposed to change decisions when a mandate is conditional on them.

The principle instead separates acquiring information, proposing or investigating a representation, approving its operational adoption, and renewing an affected delegation. An agent may record a candidate category in a nonauthoritative investigation store. It may use that candidate to explain an uncertainty or propose a review. No unadmitted semantic premise may discharge a

required permit condition. The candidate can appear as a labeled hypothesis in a review submission, without supplying operational authorization.

Version binding alone prevents accidental substitution only if the bound bundle remains the evaluated basis and the grant is current. It does not decide whether the next bundle should be adopted. Authority-relative comparison serves that admission decision, while the execution gate enforces the admitted result.

7. Building-safety thought experiment

7.1. Technical access and the decision boundary

This is a synthetic incident-management application used to illustrate an authorization problem. It does not specify a real building-safety or emergency-response procedure.

A facilities agent can monitor incidents at an assigned site, preserve observations, notify a designated responder, and close routine maintenance notices. The ontology includes `ServiceNotice`, `RoutineNotice`, and `SensorAnomaly`. An independently supported service notice can be classified under the admitted model as routine. An unexplained sensor anomaly may indicate a situation that the agent is not authorized to resolve independently. It remains within the agent's observation and reporting scope.

The technical update permission covers incident records generally. Setting a record to `resolved` can remove it from the unresolved-incident display and, for some records, trigger an all-clear message. Those effects matter to people who use the display or receive the message. The decision evaluator must examine the proposed closure and its consequences, rather than treat every successful status update as equally authorized. A recommendation also needs an accurate effect description: if downstream automation treats it as a final resolution, the recommendation label does not remove that consequence.

In the simplified policy, routine membership is the predicate defining which notices the agent may close. The other conditions check that the incident belongs to its assigned site and that the observations have independently attested provenance and completeness. These evidence checks do not certify that a possible hazard has been cleared. The distinction is essential to the following comparison: the category boundary carries the decision-scope restriction. A policy with an additional, independently checked safety-clearance requirement would retain that requirement after a taxonomy edit.

7.2. A proposed category change

An incident c has the already supported type `SensorAnomaly`, belongs to the assigned site, and has observations meeting the provenance and com-

pleteness checks. The fixed proposal is to mark that incident resolved and publish its all-clear status; it makes no claim about other incidents or the safety of the whole site. Under O_0, the accepted classification does not entail `RoutineNotice(c)`, so the policy does not permit the agent to close it. The agent can retain the unresolved status, preserve the observations, and notify the designated responder under its existing reporting authority.

An improvement process proposes the axiom

$$\mathsf{SensorAnomaly} \sqsubseteq \mathsf{RoutineNotice}.$$

Under O_1, the unchanged routine-closure policy now permits closure of c in the hypothetical evaluation. The comparison has a concrete witness: the same incident, evidence, assigned-site condition, proposed effect, and policy change from non-permit to permit under U. This is prospective eligibility expansion. It does not demonstrate that closure is safe or that the category change should be accepted. If a separate safety-clearance condition were required and remained unsatisfied, the proposal would still be refused; a subclass axiom does not override that condition.

The active grant remains bound to B_0, so V does not authorize use of the proposed category change to close the incident. The domain steward reviews whether the classification is justified. The role responsible for delegating incident-resolution authority separately decides whether the resulting scope is acceptable. A descriptive model can be accepted while authority to close incidents in the new category is withheld. Review can reject the broad subclass relation, distinguish a narrower category supported by additional evidence, or change the policy before any revised operational grant is issued. Even an explicitly approved change can be mistaken; approval and version binding do not establish that the underlying situation is safe.

An ungoverned deployment could instead replace the ontology service's current hierarchy and continue evaluating the old policy against it. That would convert a change in description into a change in the agent's effective decision scope. Pinning only the policy text, or naming an ontology `current`, would not prevent the expansion. A version-bound grant must identify the meanings against which the decision scope was evaluated.

7.3. Changed facts without a changed ontology

Suppose the ontology remains O_0 but the agent writes `RoutineNotice(c)` into the situation store. The same simplified policy can now permit closure. This is an assertion-admission problem rather than an ontology revision.

The approved admission rule must identify who can attest routine status and what evidence supports it. Permission to record a sensor observation

does not automatically include permission to classify that observation as grounds for closing the incident. A self-issued routine classification cannot become an authoritative premise merely because the agent can write to the store. New genuine routine notices can enter through the existing attestation procedure without a new ontology or a broader delegation.

The agent therefore has two distinct ways to respond to an unfamiliar situation: it can report the observation under its current authority, and it can propose a new classification for review. Neither response requires pretending that the situation is already understood or granting itself authority to resolve it.

8. Customer-support thought experiment

8.1. Access, judgment, and consequences

A support agent can access customer records and issue a courtesy credit of at most 25 units for an attested `CourtesyEligibleIssue`. Its mandate excludes contract termination, fraud classification, and modification of contractual entitlements. It can propose those outcomes to a designated reviewer.

Access to a customer's subscription and payment history does not itself confer authority to cancel that subscription or label the customer fraudulent. If the tool exposes both credit and cancellation operations, the evaluator must distinguish those outcomes. Credit limits must also consider accumulated effects: four nominally allowed credits can violate a 25-unit per-incident limit. The relevant context includes previous credits and incident identity. Entity resolution and aggregation are part of the governing basis, not optional audit details.

The agent encounters duplicate renewals caused by a migration. The current issue model has no adequate category. It records the observations and proposes `MigrationDuplicateRenewal`. It may explain the mismatch to the improvement function. Until adopted through the required process, the proposed class does not justify an autonomous credit or a change to subscription status.

8.2. Change, review, and an allowed resolution

The improvement function proposes to place `MigrationDuplicate Renewal` beneath `CourtesyEligibleIssue`. For a taxonomy-only comparison, the change assessor fixes customer/evidence snapshots that include a supported `MigrationDuplicateRenewal` classification for the affected incidents, alongside fixed credit proposals. This candidate classification is available only for shadow analysis until operationally admitted. If the classifier or assertion-admission rule also changes, the original observations

must instead be replayed through both admission pipelines, as described in Section 4.2. Credits previously refused because courtesy eligibility was not established become permitted under the new hierarchy. In addition, some duplicate charges exceed 25 units. The semantic expansion witness and amount constraint are different findings: new class membership does not remove an outcome ceiling.

A reviewer may adopt the new descriptive class while deciding that these cases require billing reconciliation rather than courtesy compensation. The domain ontology can correctly represent the issue without delegating its resolution to the support agent. Another deployment may explicitly authorize credits up to the existing ceiling, with larger corrections escalated. In either case, operational use follows a grant bound to the approved model, evidence rule, and policy.

An external ticket code mapping can cause the same problem without adding a class. If a new source code maps to courtesy eligibility, its effect on the policy query must be evaluated. An unsupervised classifier assigning courtesy status can also change current permits. All three mechanisms require control, but they need different owners and evidence.

8.3. Costs and failure cases

Conservative refusal delays some legitimate service. A narrow taxonomy can encode institutional blind spots, and an approval queue can preserve them. The architecture should therefore provide a clear unknown-category route, preserve contrary evidence, allow authorized exception decisions, and measure review burden. Bounded agency must not become a rationale for suppressing observations that challenge the model.

Nor does approval solve representation errors. A mistaken equivalence, misleading source mapping, fabricated receipt, or incomplete consequence contract can lead a deterministic evaluator to an incorrect permit. The system needs domain validation and monitoring in addition to formal consistency. Shared meanings can propagate a mistake across many agents, making governed semantic releases valuable and consequential at the same time.

9. Develop, operate, and improve

A shared semantic basis can support three functions with different grants.

Develop implements capabilities and adapters against an identified model and policy contract. Its permission to modify code or author test data does not entail permission to release production semantics, rewrite live mandates, or commit domain decisions. Development output remains a proposed artifact until the release authority admits it.

Operate interprets admitted situations and proposes or executes decisions within the active contract. It can collect observations and report mismatches. It cannot grant itself a broader decision class or make a candidate ontology extension authoritative simply because the extension appears useful.

Improve investigates failures and gaps, compares alternatives, and proposes changes. It can have a broader research scope than the operating function while retaining narrower authority over effects. Discovery of a new category, a successful test, or a persuasive diagnosis does not itself approve adoption.

An orchestrator or another authorized mechanism governs transitions between these functions. They may be different agents, services, teams, or roles in one platform; naming three agents does not establish separation of duties if they share unrestricted credentials. The full develop/operate/improve lifecycle is outside this paper's scope. Here it shows why access to a common ontology need not imply a common authority envelope.

10. Discussion and future work

10.1. Relationship to existing mechanisms

The architecture combines established semantic authorization, normative modeling, ontology-change analysis, and version-bound enforcement [1-3, 12-19, 22]. Its organizing question is: *Does a candidate semantic change alter the decisions eligible under a stated delegation and comparison domain, and which role may ratify that effect?* The comparison target makes the effect of a semantic change explicit, while the admission procedure connects that effect to a designated authority. The taxonomy example illustrates this relationship in a restricted fragment; it does not establish a general preservation theorem.

A typed schema, curated decision table, state machine, or existing policy data model may represent a narrow domain more cheaply than an ontology. The ontology proposal is appropriate when shared relations and inference make policy interpretation depend on evolving meaning. It should be compared with simpler representations rather than treated as a mandatory foundation for every agent.

10.2. Technical and institutional limits

The formal model is relative to represented proposals, evidence, and consequences. It cannot certify that the institution's mandate is legitimate, that classifications are fair, or that the admitted evidence is true. Classical ontology consistency and SHACL conformance do not establish these claims.

Complete mediation may be infeasible across legacy tools and external workflows. Direct credentials, unmodeled notification effects, and data disclosed during retrieval can bypass a final write gate. Multiagent composition creates additional problems: individually allowed proposals can combine into an unauthorized outcome. Trace and aggregate constraints need state, concurrency control, and clearly attributed effects.

The semantic comparison problem can be expensive or undecidable for sufficiently expressive languages and open domains. Extracted dependencies can be incomplete. A narrower sound approximation may refuse benign changes; a broader heuristic may miss harmful ones. An indeterminate result must remain visible, with a policy defining which authority may accept the residual uncertainty.

Institutions can also approve an overbroad grant. Separating approval roles supports accountability; it does not guarantee prudent judgment. Evidence disputes, appeal, jurisdiction, and override procedures require domain-specific governance that this paper does not supply.

10.3. What an implementation study should test

A future study should compare four configurations on the same workflows and evidence: resource/operation permission checks alone; an explicit business-decision policy gate; that gate with version-bound semantics and current grants; and that gate with authority-relative semantic-change review. The third configuration must already include competent semantic-policy administration, ontology-change analysis, and its usual approval process. The fourth should add only the explicit delegation-relative comparison and ratification procedure. Otherwise the study would attribute the benefits of basic change control to the proposed procedure.

The incremental benefit is an open hypothesis. A useful result would identify consequential changes missed by that strong baseline or reduce review burden at comparable detection quality. Equivalent decisions at equal or greater cost would count against the added procedure's utility. Domain reviewers independent of the implementation should label intended decision scope before seeing the comparison results, and report disagreements rather than treating the evaluator's own output as ground truth.

Tests should include subclass additions, equivalence and mapping changes, ordinary new instances, disputed type assertions, imported rules, reasoner changes, unrelated ontology edits, expired grants, concurrent releases, stale evidence, bypass credentials, cumulative credits, and recommendations that downstream systems finalize. Report expansion witnesses, unauthorized effects, false refusals, review workload, latency, and

recovery separately. A bounded executable example does not constitute an experiment on agent reliability or a deployment evaluation.

Future work should specify a decidable comparison fragment, implement a change assessor, validate the mandates, and compare detection accuracy and cost against strong existing semantic-policy and ontology-change controls. These comparisons should establish whether the additional review detects consequential authority expansions at acceptable cost.

11. Conclusion

Delegated intelligence needs an explicit account of which judgments and consequences an institution has entrusted to it. An independently governed shared ontology can help express that account, while ontology revisions, mappings, admitted assertions, and inference choices can also change its effective scope. A policy can remain textually identical as the decisions it permits expand.

Bounded agency, as developed here, separates acquiring or proposing knowledge from adopting an authoritative operational basis and renewing affected decision rights. The proposed architecture pairs authority-relative semantic comparison with explicit ratification, current version-bound authorization, and mediated effects. Future evaluation should test whether this obligation detects consequential semantic changes beyond existing controls and measure review burden and runtime cost.

Declarations

Nature of the work. This is a conceptual design study. The cases and finite taxonomy illustration are synthetic thought experiments. No human-subject study, production deployment, or agent-performance experiment was conducted.

Research artifacts. Manuscript sources, the executable finite-taxonomy illustration, its recorded output, and an example change-admission schema are available in the supplementary archive and project repository.

AI assistance. OpenAI Codex assisted literature discovery, manuscript drafting, generation of illustrative code, and internal critical review. Reference metadata was screened with RefChecker 3.1.6, and automated findings were reviewed against publisher and other primary-source records. The author is responsible for the manuscript's content, references, and conclusions.

References

[1] Andrzej Uszok, Jeffrey M. Bradshaw, Matthew Johnson, Renia Jeffers, Austin Tate, Jeff Dalton, and Stuart Aitken. 2004. "KAoS Policy Management for Semantic Web Services." *IEEE Intelligent Systems* 19(4): 32–41. doi:10.1109/MIS.2004.31.

[2] Lalana Kagal, Tim Finin, and Anupam Joshi. 2003. "A Policy Based Approach to Security for the Semantic Web." *ISWC*, LNCS 2870: 402–418. doi:10.1007/978-3-540-39718-2_26.

[3] Guido Boella and Leendert van der Torre. 2004. "Regulative and Constitutive Norms in Normative Multiagent Systems." *KR 2004*: 255–265. Primary AAAI paper.

[4] Vincent C. Hu, David Ferraiolo, D. Richard Kuhn, Adam Schnitzer, Kenneth Sandlin, Robert Miller, and Karen Scarfone. 2014, updated 2019. *Guide to Attribute Based Access Control (ABAC) Definition and Considerations*. NIST SP 800-162. doi:10.6028/NIST.SP.800-162.

[5] Ruoming Pang, Ramon Caceres, Mike Burrows, Zhifeng Chen, Pratik Dave, Nathan Germer, Alexander Golynski, Kevin Graney, Nina Kang, Lea Kissner, Jeffrey L. Korn, Abhishek Parmar, Christina D. Richards, and Mengzhi Wang. 2019. "Zanzibar: Google's Consistent, Global Authorization System." *USENIX ATC*: 33–46. Primary USENIX paper and record.

[6] Jaehong Park and Ravi Sandhu. 2004. "The UCONABC Usage Control Model." *ACM Transactions on Information and System Security* 7(1): 128–174. doi:10.1145/984334.984339.

[7] Joseph W. Cutler, Craig Disselkoen, Aaron Eline, Shaobo He, Kyle Headley, Michael Hicks, Kesha Hietala, Eleftherios Ioannidis, John Kastner, Anwar Mamat, Darin McAdams, Matt McCutchen, Neha Rungta, Emina Torlak, and Andrew Wells. 2024. "Cedar: A New Language for Expressive, Fast, Safe, and Analyzable Authorization (Extended Version)." arXiv:2403.04651v2. Primary record.

[8] Open Policy Agent project. Undated. *Open Policy Agent Documentation*. Official documentation. Accessed 9 October 2026.

[9] Renato Iannella and Serena Villata, eds. 2018. *ODRL Information Model 2.2*. W3C Recommendation, 15 February. Dated specification.

[10] Pascal Hitzler, Markus Krötzsch, Bijan Parsia, Peter F. Patel-Schneider, and Sebastian Rudolph, eds. 2012. *OWL 2 Web Ontology Language Primer (Second Edition)*. W3C Recommendation, 11 December. Dated specification.

[11] Holger Knublauch and Dimitris Kontokostas, eds. 2017. *Shapes Constraint Language (SHACL)*. W3C Recommendation, 20 July. Dated specification.

[12] Elena Botoeva, Boris Konev, Carsten Lutz, Vladislav Ryzhikov, Frank Wolter, and Michael Zakharyaschev. 2017. "Inseparability and Conservative Extensions of Description Logic Ontologies: A Survey." *Reasoning Web: Logical Foundation of Knowledge Graph Construction and Query Answering*, LNCS 9885: 27–89. RW2016 event; publisher publication 2017. doi:10.1007/978-3-319-49493-7_2; author-deposited text.

[13] Sven Peldszus, Jens Bürger, Timo Kehrer, and Jan Jürjens. 2021. "Ontology-driven Evolution of Software Security." *Data & Knowledge Engineering* 134: 101907. doi:10.1016/j.datak.2021.101907.

[14] Bo Nørregaard Jørgensen and Zheng Grace Ma. 2026. "Governing Agentic AI in Enterprise Workflows: A Bounded-Autonomy Framework for Delegated Authority and Controlled Execution." *Information* 17(9): 923. Published 20 September. doi:10.3390/info17090923.

[15] Anuj Kaul, Qianlong Lan, and Pranay Gupta. 2026. "Behavioral Governance for Autonomous AI Agents: The AgentBound Framework." arXiv:2606.30970v2, revised 1 July. Primary record.

[16] Parviz Shariff. 2026. "The Authority Resolution Framework: A Five-Domain Ontology for Governing Who and What Decides, at Scale." arXiv:2608.15832v1. Primary record.

[17] Yingqi Zhang. 2026. "Agent libOS: A Runtime Substrate for Capability-Controlled Self-Evolving LLM Agents." arXiv:2606.03895v3, revised 18 August. Primary record.

[18] Basavesh Ammanaghatta Shivakumar, Swarn Priya, and Peng Gao. 2026. "AgentFlow: A Flow-Centric Policy Language and Framework for Securing LLM Agent Systems." arXiv:2608.22868v1. Primary record.

[19] Jesus Salas. 2026. "Authority at Commit Time: Reject-and-Rerun Semantics for Governed Agentic Systems." arXiv:2609.31490v1. Primary record.

[20] Fred B. Schneider. 2000. "Enforceable Security Policies." *ACM Transactions on Information and System Security* 3(1): 30–50. Author-hosted primary paper.

[21] Eugene F. Fama and Michael C. Jensen. 1983. "Separation of Ownership and Control." *Journal of Law and Economics* 26(2): 301–325. doi:10.1086/467037.

[22] Inferal. Undated. *Inferal Governed Artifact Evolution Ontology.* Unofficial 1.0 Editor's Draft. Maintained specification. Accessed 9 October 2026; no dated immutable publication established.

[23] Andrzej Uszok, Jeffrey M. Bradshaw, James Lott, Maggie Breedy, Larry Bunch, Paul Feltovich, Matthew Johnson, and Hyuckchul Jung. 2008. "New Developments in Ontology-Based Policy Management: Increasing the Practicality and Comprehensiveness of KAoS." *IEEE POLICY*: 145–152. doi:10.1109/POLICY.2008.47; primary manuscript.

[24] Nuno Lopes, Sabrina Kirrane, Antoine Zimmermann, Axel Polleres, and Alessandra Mileo. 2012. "A Logic Programming Approach for Access Control over RDF." *ICLP Technical Communications*, LIPIcs 17: 381–392. doi:10.4230/LIPIcs.ICLP.2012.381.

[25] Andrew J. I. Jones and Marek Sergot. 1996. "A Formal Characterisation of Institutionalised Power." *Logic Journal of the IGPL* 4(3): 427–443. doi:10.1093/jigpal/4.3.427.

[26] Tobin South, Samuele Marro, Thomas Hardjono, Robert Mahari, Cedric Deslandes Whitney, Dazza Greenwood, Alan Chan, and Alex Pentland. 2025. "Authenticated Delegation and Authorized AI Agents." arXiv:2501.09674v1. Primary record.

[27] Nenad Tomašev, Matija Franklin, and Simon Osindero. 2026. "Intelligent AI Delegation." arXiv:2602.11865v1. Primary record.

[28] Lalana Kagal. Undated. *Rei Ontology Specifications, Ver 2.0.* Author-maintained specification. Accessed 9 October 2026.